Chapter 1: The Fundamental Unit of Life

While examining a thin slice of cork, Robert Hooke saw that the cork resembled the structure of a honeycomb consisting of many little compartments. Hooke called these boxes cells. A cell is the basic structural and functional unit of life. The plasma membrane is the outermost covering of the cell. It separates the contents of the cell from its external environment and permits the entry and exit of some materials. The cell membrane is therefore called a selectively permeable membrane. Diffusion is the spontaneous movement of a substance from a region of high concentration to a region where its concentration is low. Osmosis is the movement of water across a selectively permeable membrane.